

Problem. Let Γ_1 and Γ_2 be two distinct circles with centers at O_1 and O_2 and radii r_1 and r_2 , and let $d = O_1O_2$. Show that Γ_1 is tangent to Γ_2 if and only if

$$d = r_1 + r_2 \quad \text{or} \quad d = |r_1 - r_2|.$$

Let P be the common point of the circles Γ_1 and Γ_2 .

WLOG assume $r_2 \geq r_1$.

Since the circles are tangent
NO! $\angle O_1 P O_2 = 90^\circ = 180^\circ!$

By the Pythagorean theorem

$$d^2 = r_1^2 + r_2^2 \quad \text{and}$$

$$r_2^2 = d^2 - r_1^2 = (d+r_1)(d-r_1).$$

Therefore, either NO!

$$r_2 = d + r_1 \quad \text{or} \quad r_2 = d - r_1.$$

Hence the statement follows.